

Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1  
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.2.3
```

```
## Warning: package 'ggplot2' was built under R version 4.2.3
```

```
## Warning: package 'tibble' was built under R version 4.2.3
```

```
## Warning: package 'tidyr' was built under R version 4.2.3
```

```
## Warning: package 'readr' was built under R version 4.2.3
```

```
## Warning: package 'purrr' was built under R version 4.2.3
```

```
## Warning: package 'dplyr' was built under R version 4.2.3
```

```
## Warning: package 'stringr' was built under R version 4.2.3
```

```
## Warning: package 'forcats' was built under R version 4.2.3
```

```
## Warning: package 'lubridate' was built under R version 4.2.3
```

```
library(rvest)
```

```
## Warning: package 'rvest' was built under R version 4.2.3
```

```
getwd()
```

```
## [1] "C:/Users/zhc/Desktop/R/EDE_Spring2026/Assignments"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
- Set the year to 2024
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2
url <- "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024"
the_website <- read_html(url)
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
 - Water system name
 - Contact Person
 - Ownership
- From the “3. Water Supply Sources” section:
 - Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “Reginald Hicks”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

#3

```
the_water_system_name_tag <- "div+ table tr:nth-child(1) td:nth-child(2)"
the_contact_person_tag <- "div+ table tr:nth-child(4) td:nth-child(2)"
the_ownership_tag <- "div+ table tr:nth-child(2) td:nth-child(4)"
the_max_day_use_tag <- "th~ td+ td"
```

```
water_system_name <- the_website %>% html_nodes(the_water_system_name_tag) %>% html_text(trim = TRUE) %>%
contact_person <- the_website %>% html_nodes(the_contact_person_tag) %>% html_text(trim = TRUE) %>% .[1]
ownership <- the_website %>% html_nodes(the_ownership_tag) %>% html_text(trim = TRUE) %>% .[1]
max_day_use <- the_website %>% html_nodes(the_max_day_use_tag) %>% html_text(trim = TRUE)
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

#4

```
df_withdrawals <- data.frame(
  Month = c("Jan", "May", "Sept", "Feb", "Jun", "Oct", "Mar", "Jul", "Nov", "Apr", "Aug", "Dec"),
  Year = rep(2024, 12),
  WaterSystemName = rep(water_system_name, 12),
  ContactPerson = rep(contact_person, 12),
  Ownership = rep(ownership, 12),
  MaxDayUse_MGD = as.numeric(max_day_use),
  stringsAsFactors = FALSE
)
```

```
df_withdrawals$Date <- my(paste(df_withdrawals$Month, "-", df_withdrawals$Year))
```

```
df_withdrawals <- df_withdrawals[order(df_withdrawals$Date), ]
```

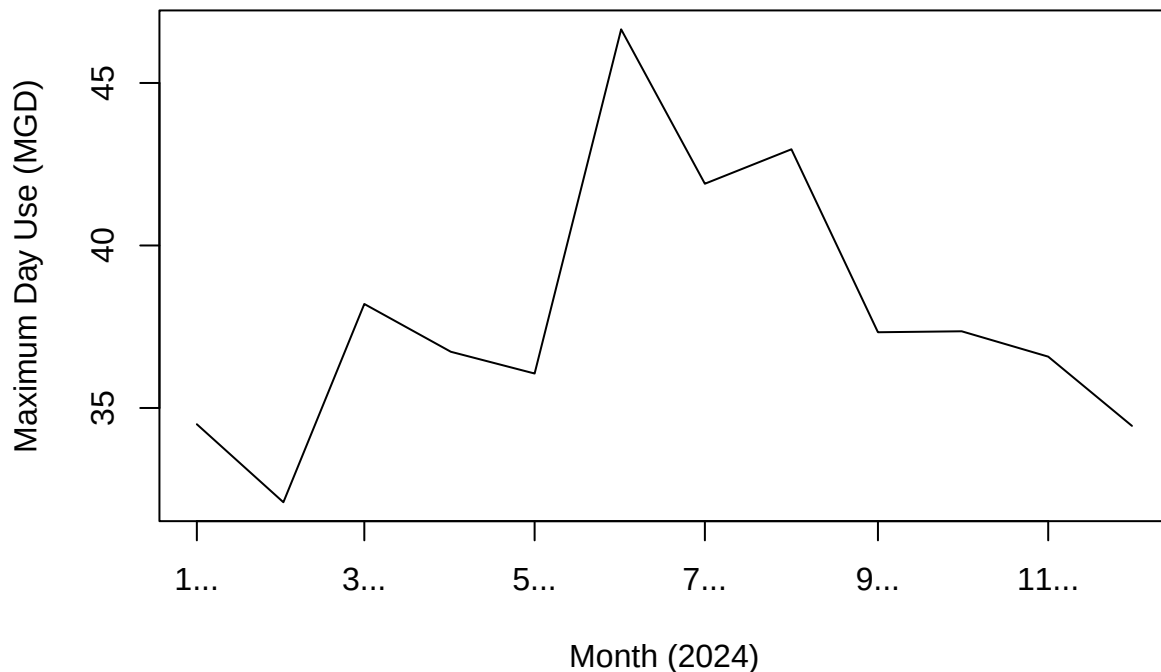
```
View(df_withdrawals)
```

#5

```
plot(df_withdrawals$Date, df_withdrawals$MaxDayUse_MGD, type = "l",
     xlab = "Month (2024)", ylab = "Maximum Day Use (MGD)")
```

```
## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '1' in 'mbcsToSbcs': dot substituted for <e6>
```

```
## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '1' in 'mbcsToSbcs': dot substituted for <9c>
```

6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs, “PWSID” and “year”, that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

#6.

```
scrape_lwsp <- function(PWSID, the_year){
  the_scrape_url <- paste0(
    "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=",
    PWSID,
    "&year=",
    the_year
  )
  the_website <- read_html(the_scrape_url)
  the_water_system_name_tag <- "div+ table tr:nth-child(1) td:nth-child(2)"
  the_contact_person_tag <- "div+ table tr:nth-child(4) td:nth-child(2)"
  the_ownership_tag <- "div+ table tr:nth-child(2) td:nth-child(4)"
  the_data_tag <- "th~ td+ td"
  water_system_name <- the_website %>% html_nodes(the_water_system_name_tag) %>% html_text(trim = TRUE)
  contact_person <- the_website %>% html_nodes(the_contact_person_tag) %>% html_text(trim = TRUE)
  ownership <- the_website %>% html_nodes(the_ownership_tag) %>% html_text(trim = TRUE)
```

```

max_day_use      <- the_website %>% html_nodes(the_data_tag) %>% html_text(trim = TRUE)
df_withdrawals <- data.frame(
  Month = c("Jan", "May", "Sept", "Feb", "Jun", "Oct", "Mar", "Jul", "Nov", "Apr", "Aug", "Dec"),
  Year  = rep(the_year, 12),
  PWSID = rep(PWSID, 12),
  MaxDayUse_MGD = as.numeric(max_day_use[1:12])
)
df_withdrawals <- df_withdrawals %>%
  mutate(
    WaterSystemName = !!water_system_name,
    ContactPerson    = !!contact_person,
    Ownership        = !!ownership,
    Date = my(paste(Month, "-", Year))
  )

df_withdrawals <- df_withdrawals[order(df_withdrawals$Date), ]

return(df_withdrawals)}
df1 <- scrape_lwsp("03-32-010", 2023)
View(df1)

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```

#7
df_2020 <- scrape_lwsp("03-32-010", 2020)

plot(df_2020$Date, df_2020$MaxDayUse_MGD, type = "l",
     xlab = "Month (2020)", ylab = "Maximum Day Use (MGD)")

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '1 ' in 'mbcsToSbcs': dot substituted for <e6>

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '1 ' in 'mbcsToSbcs': dot substituted for <9c>

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '1 ' in 'mbcsToSbcs': dot substituted for <88>

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '1 ' in 'mbcsToSbcs': dot substituted for <e6>

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '1 ' in 'mbcsToSbcs': dot substituted for <9c>

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '1 ' in 'mbcsToSbcs': dot substituted for <88>

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '3 ' in 'mbcsToSbcs': dot substituted for <e6>

```



```

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '9 ' in 'mbcsToSbcs': dot substituted for <e6>

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '9 ' in 'mbcsToSbcs': dot substituted for <9c>

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '9 ' in 'mbcsToSbcs': dot substituted for <88>

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '9 ' in 'mbcsToSbcs': dot substituted for <e6>

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
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## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '9 ' in 'mbcsToSbcs': dot substituted for <88>

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '11 ' in 'mbcsToSbcs': dot substituted for <e6>

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '11 ' in 'mbcsToSbcs': dot substituted for <9c>

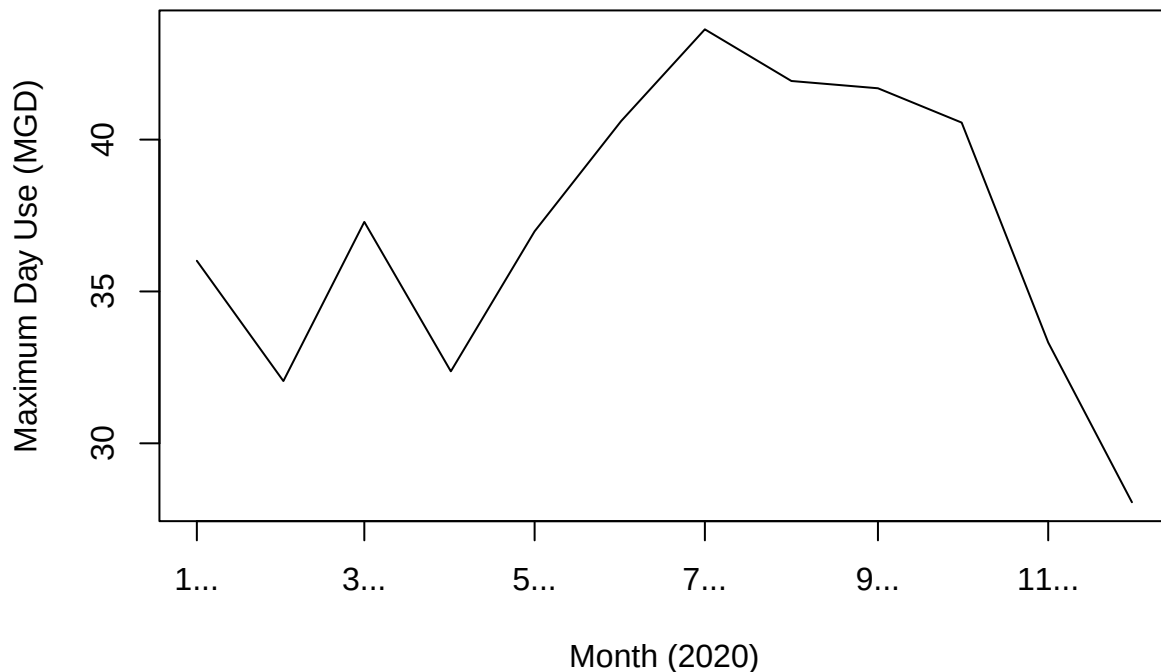
## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '11 ' in 'mbcsToSbcs': dot substituted for <88>

## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
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## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '11 ' in 'mbcsToSbcs': dot substituted for <88>

```



```
str(df_2020)
```

```
## 'data.frame':  12 obs. of  8 variables:
## $ Month      : chr  "Jan" "Feb" "Mar" "Apr" ...
## $ Year       : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ PWSID      : chr  "03-32-010" "03-32-010" "03-32-010" "03-32-010" ...
## $ MaxDayUse_MGD : num  36 32 37.3 32.4 37 ...
## $ WaterSystemName: chr  "Durham" "Durham" "Durham" "Durham" ...
## $ ContactPerson : chr  "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" ...
## $ Ownership   : chr  "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ Date       : Date, format: "2020-01-01" "2020-02-01" ...
```

8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8
# Cary 2020
df_cary_2020 <- scrape_lwsp("03-92-020", 2020)

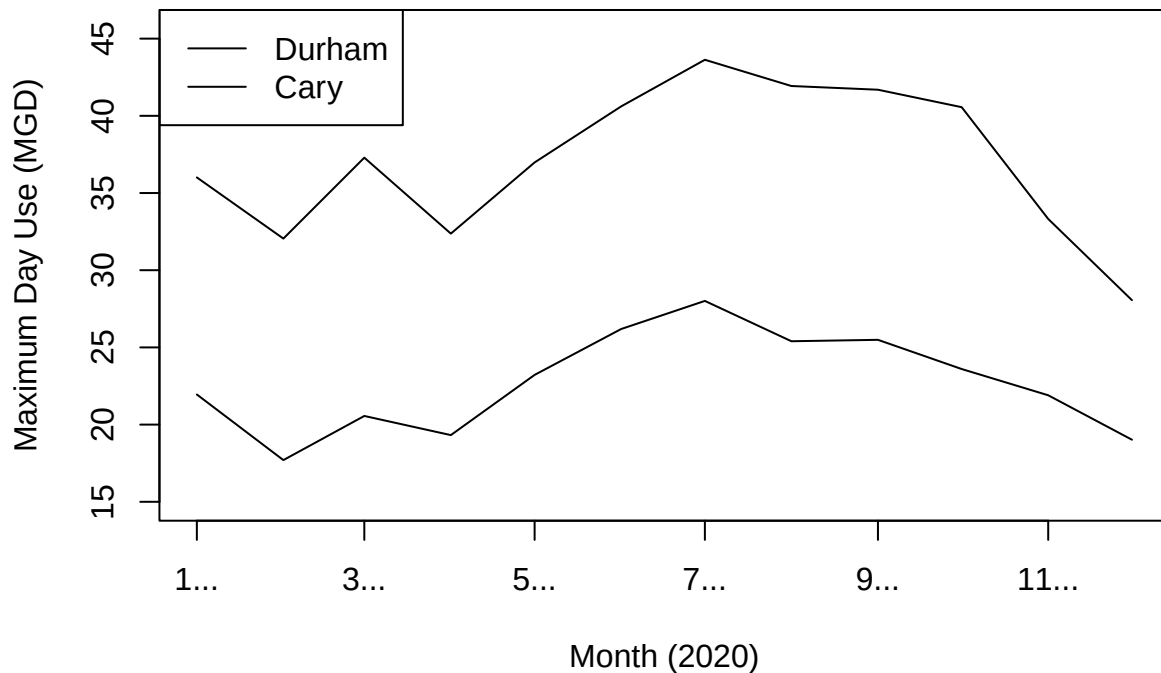
# Combine with Durham 2020 (df_2020 from Q7)
df_both_2020 <- rbind(df_2020, df_cary_2020)
plot(df_2020$Date, df_2020$MaxDayUse_MGD, type = "l",
     ylim = c(15, max(c(df_2020$MaxDayUse_MGD, df_cary_2020$MaxDayUse_MGD), na.rm = TRUE) + 2),
     xlab = "Month (2020)", ylab = "Maximum Day Use (MGD)")
```

[illegible]


```
## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '11' in 'mbcsToSbcs': dot substituted for <9c>
```

```
## Warning in axis(side, at = z, labels = labels, ...): conversion failure on
## '11' in 'mbcsToSbcs': dot substituted for <88>
```

```
lines(df_cary_2020$Date, df_cary_2020$MaxDayUse_MGD)
legend("topleft", legend = c("Durham", "Cary"), lty = 1)
```



- Use the code & function you created above to plot Cary’s max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = ‘loess’).

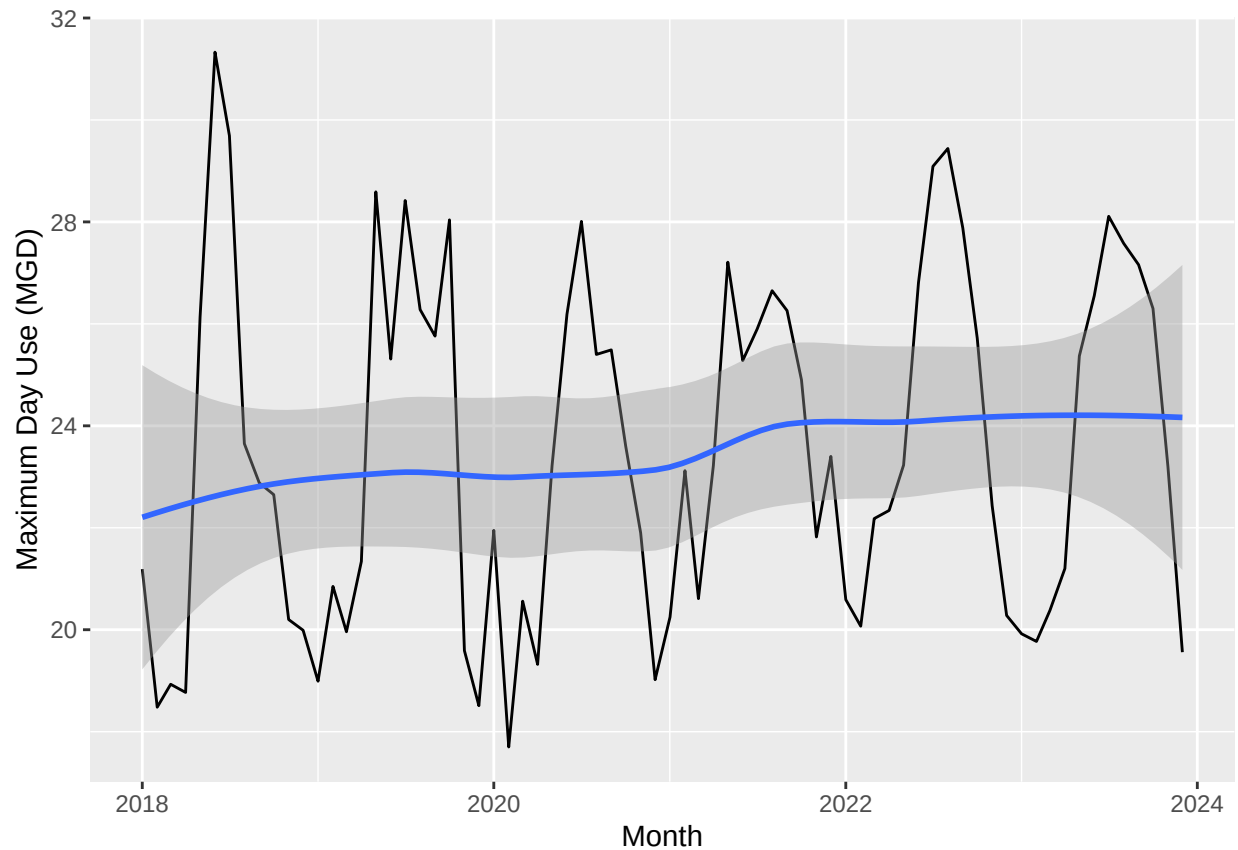
TIP: See Section 3.2 in the “06_Data_Scraping.Rmd” where we apply “map2()” to iteratively run a function over two inputs (where one of the two inputs does not change). Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9
years <- 2018:2023

df_cary_all <- map2_dfr(rep("03-92-020", length(years)), years, scrape_lwsp)

ggplot(df_cary_all, aes(x = Date, y = MaxDayUse_MGD)) +
  geom_line() +
  geom_smooth(method = "loess") +
  labs(x = "Month", y = "Maximum Day Use (MGD)")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: Yes, it looks like Cary's max daily water use shows a slight upward trend from 2018 to 2023. The monthly values bounce around a lot, but the smoothed line gradually increases over time and then levels off a bit near the end. >